

Demo of PortaChrome: A Portable Contact Light Source for Integrated Re-Programmable Multi-Color Textures

Yunyi Zhu
MIT CSAIL
Cambridge, USA
yunyizhu@mit.edu

Cedric Honnet
MIT
Cambridge, USA
honnet@mit.edu

Yixiao Kang
UC Berkeley
Berkeley, USA
yixiao_kang@berkeley.edu

Junyi Zhu
MIT CSAIL
Cambridge, USA
junyizhu@mit.edu

Angelina J. Zheng
MIT CSAIL
Cambridge, USA
ajz@mit.edu

Kyle Heinz
MIT CSAIL
Cambridge, USA
kyleh51@mit.edu

Grace Tang
MIT CSAIL
Cambridge, USA
gtang@mit.edu

Luca Musk
MIT CSAIL
Cambridge, USA
lulew@mit.edu

Michael Wessely
Aarhus University
Aarhus, Denmark
michael.wessely@cs.au.dk

Stefanie Mueller
MIT CSAIL
Cambridge, USA
stefanie.mueller@mit.edu

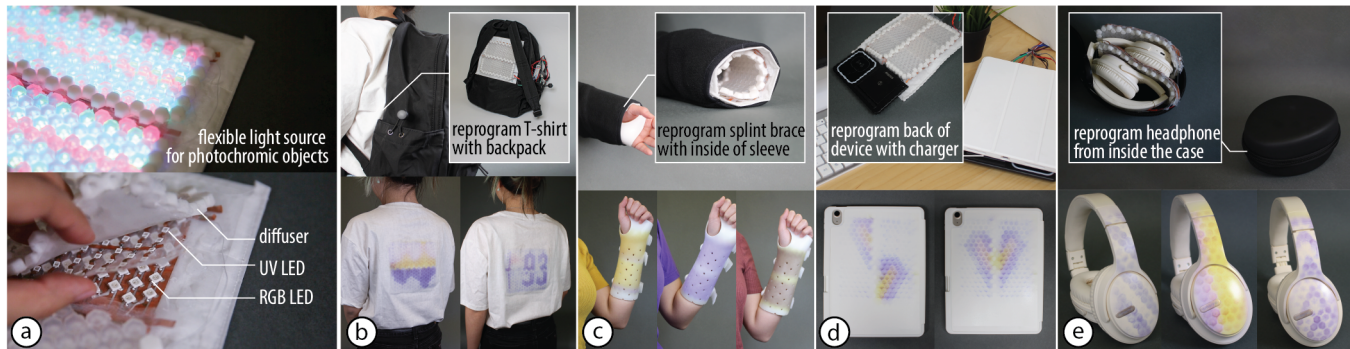


Figure 1: (a) PortaChrome is a flexible contact light source that can reprogram the appearance of an object previously coated with photochromic material. PortaChrome has integrated UV and RGB LEDs that saturate and desaturate the color channels of photochromic material. PortaChrome is portable and flexible, allowing it to be attached to (b) a backpack to reprogram the back of clothes, (c) the inside of a sleeve to reprogram a splint brace, (d) a wireless charger to reprogram the back of a device, and (e) the inside of a headphone case to reprogram the design of a headphone.

ABSTRACT

In this demo, we present *PortaChrome*, a portable light source that can be attached to everyday objects to reprogram the color of surfaces in contact with them. When *PortaChrome* makes contact with objects that were previously coated with photochromic dyes, the UV and RGBs LEDs inside *PortaChrome* create multi-color textures on the objects. In contrast to prior work, which used projectors for the color-change, *PortaChrome* has a thin and flexible form factor, which allows the color-change process to be integrated into daily user interactions. Because of the close distance between the light

source and the photochromic object, *PortaChrome* creates color textures in less than 4 minutes on average, which is 8 times faster than prior work. We demonstrate *PortaChrome* with four application examples, including data visualizations on textiles and personalized wearables.

CCS CONCEPTS

• Human-centered computing → Human computer interaction (HCI).

KEYWORDS

digital fabrication; programmable textures; photochromic dyes

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1 INTRODUCTION

An important step towards achieving the vision of "The Ultimate Display" [8] is to augment the surfaces of objects with display capabilities through novel display fabrication methods [2]. This capability has created new application opportunities, including the ability to fit multiple designs on personal items such as phone cases and shoes [4], or to place relevant information in the daily environment of users, such as showing a calendar on a mug [9].

Researchers have developed different fabrication methods to achieve display capabilities, such as using electrochromic [3] and thermochromic [1, 5] dyes, which enable the transition between pre-defined color phases. Researchers have recently shown how to expand the color gamut using photochromic dyes (Photo-Chromeleon [4]). By spraying a mixture of cyan, magenta, and yellow photochromic dyes on an object and projecting light patterns of specific wavelengths to saturate and desaturate individual color channels, the resulting photochromic object achieves high-resolution multi-color textures that are fully reprogrammable.

One limitation of a projector system for re-programmable surfaces is that it has to be set up at a distance from the object to induce color change, limiting the types of interactions that can leverage texture reprogramming. More portable light sources enable new ways to interact with photochromic material. For example, researchers of photochromic systems have considered portable light sources to integrate color-change into users' everyday interaction, such as integrating a UV LED matrix into shoes to allow reprogramming the carpet while walking (*PhotochromicCarpet* [6]).

However, the portable light sources used in photochromic systems so far have only supported single-color textures. To enable multi-color textures, light sources do not only require UV LEDs for saturation but also RGB LEDs for individually desaturating each color channel. In addition, the portable light source used in existing photochromic systems has been rigid (e.g., rigid UV LED matrices) and thus limited the color change to flat surfaces. To enable multi-color textures on curved surfaces, such as on the curved headpiece of a pair of headphones, a flexible light source that can conform to the curved surface and has both RGB and UV wavelengths is needed. Existing flexible displays, such as commercial O-LED displays, contain RGB wavelengths, which are necessary to desaturate the photochromic dyes, but are missing the UV wavelength required for saturation.

In this demo, we present *PortaChrome*: a flexible and portable light source that supports re-programmable multi-color texture on photochromic objects (Figure 1). With a thickness of 6mm, *PortaChrome* can be embedded into surfaces that are in contact with photochromic objects, integrating the texture reprogramming process into the user's everyday interactions. We demonstrate applications such as visualizing the user's heart rate on the back of a T-shirt the user is wearing while hiking with a *PortaChrome*-augmented backpack (Figure 1b), matching the color of a splint brace with the user's everyday outfit with a *PortaChrome* embedded inside the user's sleeves (Figure 1c), displaying exercise progress with

PortaChrome attached to a wireless charger (Figure 1d), and switching the design of a headphone with *PortaChrome* integrated into the headphone casing (Figure 1e). We accomplish this by making flexible panels of RGB and UV LEDs on a textile substrate and using a silicone structure to distribute the light evenly on the surface of the object in contact. Our system also reprograms the color of an object in under 4 minutes on average, which is 8 times faster than prior work.

2 PORTACHROME

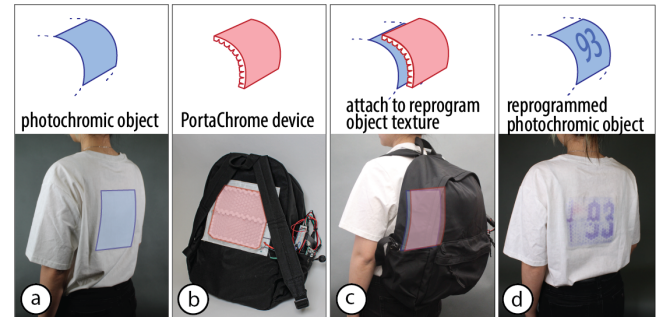


Figure 2: *PortaChrome* integrates the texture reprogramming process into everyday user interaction: (a) an object treated with photochromic material; (b) *PortaChrome* can be mounted on the surface of another object; (c) *PortaChrome* reprograms the photochromic object while being in contact with it; (d) the photochromic object can be reprogrammed into displaying different patterns.

PortaChrome can be embedded into everyday objects, thereby integrating the texture reprogramming process into the user's everyday interactions. *PortaChrome* is designed to be embedded into an object and then reprogram the color of another object previously coated with photochromic dye through direct contact. Figure 2 shows this in more detail, i.e. in Figure 2a an object is treated with photochromic dye, allowing its color to be later reprogrammed with *PortaChrome*. This object can be the back of a T-shirt, a splint brace, the back of a tablet and a headphone, as illustrated in the application scenarios. *PortaChrome* can then be attached to another object that will be in contact with the photochromic object (Figure 2b), such as a backpack, the inside of a sleeve, a wireless charger, and the inside of the casing. When in contact, *PortaChrome* illuminates UV and RGB light that reprogram the texture of the photochromic object (Figure 2c), allowing the object to display a different texture after programming is completed (Figure 2d).

Unlike previous works that use projectors for multi-color photochromic texture, *PortaChrome* enables an integrated color-changing process that does not require the user to actively design and initiate the reprogramming. This enables new interactions to create reprogrammable textures, such as data visualization that automatically update based on sensor data.

3 APPLICATIONS

We showcase applications of *PortaChrome* with the example of a data visualization on a textile surface and a personal device, and

a wrist splint and a headphone with reprogrammable customized visual appearance.

3.1 Visualize Sensor Data on Textiles with Backpack-Wearing

In our first application, we show how PortaChrome can be used to create a data visualization on a textile surface (Figure 3). In this example, we mount PortaChrome on the inside of a backpack. When the user wears the backpack over a T-shirt previously coated with photochromic dye, the backpack updates the color pattern on the T-shirt while the user wears it.



Figure 3: PortaChrome can be attached to a backpack and create customized patterns on the back of a user's T-shirt. (a) PortaChrome is sewed onto a backpack powered with a battery pack. (b) PortaChrome reprograms the back of the T-shirt to visualize the user's hiking trail and heart rate, and displays the number of active minutes spent on this hike.

In this specific example, we create a data visualization of the user's hiking experience. An altitude sensor and a heart rate sensor send data to the PortaChrome control system, which plots the altitude and heart rate as a data visualization on the T-shirt. Such a portable, contact-based application was not achievable with previous systems, which required projector setups. It is also enabled by PortaChrome's fast color change, which only took 3 minutes and 52 seconds for the hiking visualization.

3.2 Visualize Sensor Data on Back of Device during Wireless Charging

When electronic devices are wirelessly charged, their back surfaces are in contact with the wireless charging station. In this application, we demonstrate that the PortaChrome light source can be embedded

into a wireless charging station, which allows it to reprogram the back of the device that is being charged. Figure 4 shows this with an example of a tablet that has its back coated with photochromic dye. In this example, we display the user's progress toward their weekly workout challenge, where two halves of a heart move closer together and eventually align as the user reaches their weekly workout goal. In this application, each reprogramming step takes only 2 minutes and 21 seconds to complete.

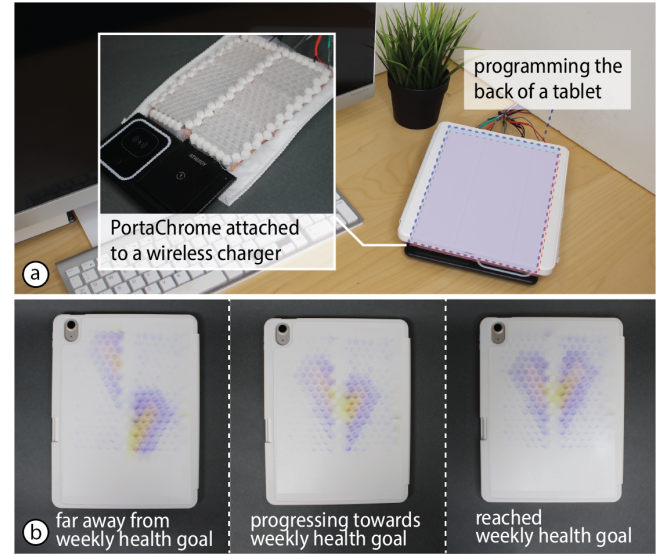


Figure 4: (a) PortaChrome can be attached to a wireless tablet charger and allows the back of the tablet to be reprogrammed while it is charging. (b) The back of this tablet case is reprogrammed to inform the user about their progress toward their weekly workout challenge.

3.3 Reprogram a Wrist Splint through Jacket-Wearing

PortaChrome can also be used to visually customize rehabilitation devices, such as wrist splints, which has been shown to increase the adoption of such devices [7]. We demonstrate this concept with the example of a wrist splint, the color of which can be reprogrammed to match the wearer's daily outfit (Figure 5). By integrating PortaChrome into the sleeves of a jacket, the user can reprogram the texture while wearing the jacket over it.

3.4 Customize Headphone in the Headphone Casing

The flexible and portable form factor allows PortaChrome to be integrated into objects' casings, such as the case of a headphone. This allows the user to reprogram the appearance of their headphones by keeping them in their casing. Figure 6 shows three different designs that we applied to the headphone by placing it into its case.



Figure 5: (a) PortaChrome can be attached to the inside of the sleeves to (b) reprogram the user's wrist splint so that it matches the user's daily outfit.



Figure 6: (a) Reprogramming headphone appearances by integrating PortaChrome with the original headphone case. (b) Three headphone designs were achieved with PortaChrome.

4 CONCLUSION

In this demo, we presented PortaChrome, a flexible and portable contact light source for reprogrammable multi-color textures on photochromic objects. Because of its flexibility and portability, PortaChrome can be placed in direct contact with the color-changing surface, allowing the light source to be attached to everyday objects and the color-changing process to be embedded in everyday user interaction. We demonstrate the new capabilities with four application examples, including data visualizations on textiles and dynamic designs on wearables.

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